

Baorian Nuchged
baorian@utexas.edu | <https://boroooo.github.io/>
Linguistics PhD student

Research Interests

Speech processing, neurolinguistics, phonetics, and information-theoretic approaches to language.

Education

University of Texas at Austin Linguistics - Neurolinguistics & Phonetics, PhD Program	Aug 2023 - present Austin, Texas
Beijing Language and Culture University Language Intelligence and Technology Master (with merit) Thesis: Quantifying information loss in second language speech using information theory	Sep 2020 - Jul 2023 Beijing, China
East China Normal University English Bachelor Thesis: A Phonological Study on Chinese Loanwords in Mongolian (Outstanding Graduation Dissertation Award)	Aug 2015 - May 2019 Shanghai

Research Experience

Graduate Research Assistant at the University of Texas at Austin Advised by Dr. Megan Crowhurst I'm working on the acoustic effects on rhythmic grouping of different languages.	May 2026 – Aug 2026
Graduate Student Researcher at Neurolinguistics Lab at the University of Texas at Austin Advised by Dr Fernando Llanos I'm working on the mechanisms of auditory information processing in the human brain, focusing on uncovering the compressed sensing processes and examining how they differ from those found in machine processing such as uniform sampling and the Nyquist theorem.	Aug 2023 - Present
Graduate Research Assistant at the University of Texas at Austin Advised by Dr. Scott Myers I am working on the acoustic analysis of Luganda H-tone syllables, to find the f0 timing and its relationship with the syllable structure.	Aug 2024 – Dec 2025
Research Assistant at Speech Acquisition and Intelligent Technology Lab (SAIT) Dr. Zhang's lab studies the speech acquisition of second-language learners and hearing-impaired children. I mainly worked on the human perception of second-language speech. -- Responsibilities include conducting perception experiments, speech data annotation, processing, and management.	Aug 2020 - Jun 2023
Research Assistant at the Department of Language and Literature of Ethnic Minorities of China Prof. Bao's department focuses on Mongolian phonetics and phonology. -- Responsibilities include data analysis, dialect speech recording, and academic paper writing.	Jun 2021 - Sep 2021

Funded Project: A Phonological Study on Chinese Loanwords in Mongolian
[Undergrad University]

Sep 2018 - Jun 2019

Student project funded by ECNU undergraduate research funding
-- Corpus analysis on Mongolian Loanword, completed by myself.

Teaching Experience

Fall 2026 Assistant Instructor

Introduction to the Study of Language

Department of Linguistics, University of Texas at Austin

Spring 2026 Guest Lecturer

Phonetics: Production and Perception of Speech Sound

Department of Linguistics, University of Texas at Austin

Spring 2026 Teaching Assistant

Phonetics: Production and Perception of Speech Sound (Instructor: Scott Myers)

Department of Linguistics, University of Texas at Austin

Fall 2024 Guest Lecturer

Introduction to the Study of Language

Department of Linguistics, University of Texas at Austin

Publications and Presentations

- **Nuchged, Baorian**, Christian Stilp, Keith Kluender, and Fernando Llanos. "Changes in narrowband spectral energy predict cortical encoding of continuous speech." under review.
- **Nuchged, Baorian**, Christian Stilp, Keith Kluender, and Fernando Llanos. "Cochlea-Scaled Entropy Predicts Robust Neural Coding of Speech Envelopes." *Seventeenth Annual Meeting of the Society for the Neurobiology of Language (SNL)*, 12–14 Sept. 2025, Gallaudet University, Washington, D.C. Poster.
- **Nuchged, Baorian**, and Rajka Smiljanic. "Protective face masks affect clearly produced diphthongs by L1 and L2 English talkers." *The Journal of the Acoustical Society of America* 155.3_Supplement (2024): A315-A315.
- **Rian Bao**, Linkai Peng, Yuchen Yan, Jinsong Zhang. "An Exploratory Study for Quantifying the Contextual Information for Successful Chinese L2 Speech Comprehension." *The 13th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, 2022. DOI: 10.1109/ISCSLP57327.2022.10038037
- Gao, Yingming, **Nuchged Baorian**, Li, Ya, and Peng, Linkai. "An Investigation of Applying Large Language Models to Spoken Language Learning." *Applied Sciences* 14.1 (2023): 224. DOI: <https://doi.org/10.3390/app14010224>
- **Rian Bao**, Linkai Peng, Yingming Gao, Jinsong Zhang. "The Contribution of Phonological and Fluency Factors to Chinese L2 Comprehensibility Ratings: A Case Study of Urdu-speaking Learners." *The 13th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, 2022. DOI: 10.1109/ISCSLP57327.2022.10038233

HONORS & AWARDS

UT Austin Professional Development Award (\$1200)

2024, 2025

Graduate Student Fellowship at University of Texas at Austin (\$25,000)	Aug 2023
Outstanding Master's Thesis Award of Beijing Language and Culture University	Jun 2023
Merit Student of Beijing Language and Culture University	Dec 2021
Best Student Paper Award in Phonetics Conference of China	Jul 2021
BLCU graduate student scholarship (¥6,000 CNY yearly)	Sep 2020, Sep 2021 and Sep 2022
BLCU graduate student research fund (¥2,000 CNY)	2022
Outstanding Bachelor's Thesis Award of East China Normal University	Jul 2019
ECNU undergraduate research fund (¥2,000 CNY)	2018
Thesis: A Phonological Study on Chinese Loanwords in Mongolian (Outstanding Graduation Thesis Award)	

WORK EXPERIENCE

Nio	Feb 2023 - May 2023
Data Specialist (Intern)	
Organizing and analyzing the wakeup speech and text data of the voice assistant Nomi. I gained experience in data management from this internship.	Beijing
ByteDance (Tiktok)	Mar 2022 - Jul 2022
English Text-to-Speech Linguistic program manager (Intern)	Beijing
Led the design and implementation of highly human-like Text-to-Speech (TTS) projects Ava and Henry (click to download demo).	
Project link: https://www.volcengine.com/docs/6561/97465#%E5%A4%9A%E8%AF%AD%E7%A7%8D	
Examining virtual sounds to make them perceptually indistinguishable from real human speech based on user feedback.	
DataBaker	Oct 2021 - Dec 2021
Speech Data Annotator (Intern)	Beijing

SKILLS LIST

Programming Languages: Python scripting (Fluent), Praat scripting (Fluent), R scripting for statistical analysis, Matlab scripting, knowledge of the Linux operating system

Data Analysis: R (ggplot2, lm, lme4, olsrr)

Experimental Design: Finding Five, Gorilla

Writing and Communication: Microsoft Office, PowerPoint, Excel, LaTeX

Languages: Mongolian (native), Chinese Mandarin (native), English (fluent), Spanish (entry-level)